

The Real-Time Frontier: Stylized Feature Quality Storytelling in Unreal Engine

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ABSTRACT

This paper presents Steamroller Animation’s innovative method for merging 2D artistic styling within a 3D real-time environment for “Spice Frontier”, leveraging Unreal Engine for streamlined long-form production. Focusing on a novel shot setup process that addresses the challenges of transitioning to a real-time workflow. We introduce solutions like custom character setups, rim shaders, rig versioning, gobos and FX sprite sheets. This approach not only aligns artistic vision with advanced technology but also advances animation workflow standards, emphasizing efficiency and creative integrity.

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1 INTRODUCTION

Steamroller Animation harnesses Unreal Engine’s real-time technology to revolutionize the animation production process, significantly shortening the timeline from concept to visualization. This advancement necessitates a novel hybrid approach, seamlessly blending the strengths of traditional animation with the efficiencies of real-time workflows. At the heart of our approach lies the ‘Shot is King’ philosophy, which elevates the importance of treating each shot as an independent unit, focusing on its autonomy and refinement. This philosophy is further supported by mechanisms for higher-level creative overrides (figure 1), ensuring that while each shot benefits from individual optimization, the collective artistic and project’s narrative vision remains cohesive and engaging.

Our innovative framework introduces tools and methodologies that leverage Unreal Engine. A cornerstone of our creative breakthrough has been the successful integration of classic 2D aesthetics within this 3D real-time setting. This not only preserves the artistic vision but also establishes a new standard for animation workflows, showcasing the powerful synergy of technological innovation and creative integrity.

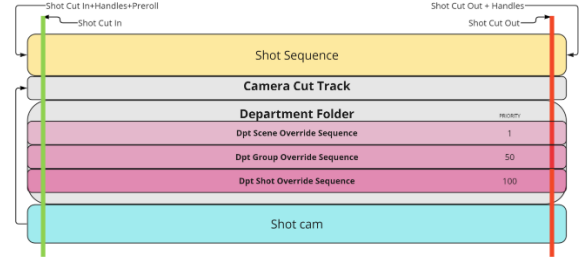


Figure 1: ‘Shot is King’ override structure with priority settings

2 SHOT SETUP PROCESS

At the core of Steamroller Animation’s approach in “Spice Frontier” is our internal ‘Shot is King’ philosophy. This principle is pivotal, redefining the traditional top-down hierarchies, where typically, a shot sequence is nested within various layers of scene sequences in a top-down manner. Instead advocating for a bottom-up approach, we prioritize constructing each shot as an independent unit. This addresses practical challenges head-on, particularly in optimizing memory usage within Unreal Engine’s real-time environment. By ensuring that each shot contains only the essential elements needed for its execution, we significantly enhance GPU memory utilization. Properly aligning the collective elements of the scene with the GPU’s capacity is key for handling complex scene management effectively. This method not only accelerates the feedback loop for artists but also guarantees smoother operation, quicker load times, and optimized rendering, essential for maintaining creative flow and productivity.

To sustain this shot-centric ethos without compromising the ability to enact broad changes, we’ve instituted a sophisticated hierarchical override system through the implementation of subsequence overrides. Scene overrides are widely recognized for their ability to enforce overarching changes. Additionally, we introduce ‘master’ overrides. Masters represent groups of shots within a scene that share similarities, such as environmental similarities or camera angles. These ‘masters’ are dynamically defined by supervisors and the production team based on specific needs, allowing for targeted refinements within these subsets. At the granular level, individual shots can then receive tailored adjustments. By leveraging Unreal Engine’s Sequencer and its built-in Hierarchical Bias feature, we ensure that Scene, Master, and Shot changes are applied in a clear hierarchical manner.

By structuring shots to accommodate scene, master, and shot-level subsequences for various departments, we enable parallel workflows across multiple departments (Figure 2a). Unlike traditional methods that require downstream handoffs or lengthy

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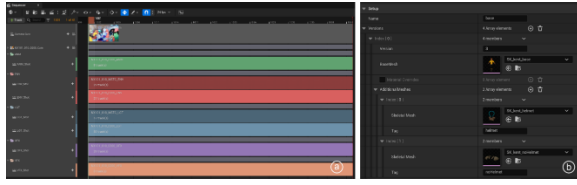


Figure 2: a) Unreal sequence structure showing override setup and 'Shot is King' ethos b) Character controls utilized to create 2D aesthetics

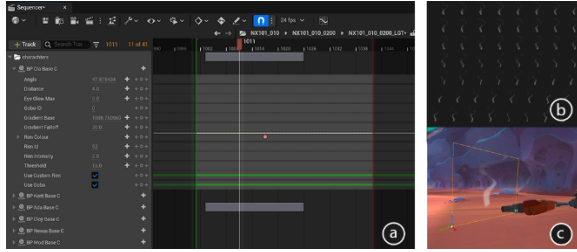


Figure 3: (a) Character rim and gobo properties exposed in sequencer for artist control (b) 2D image sequence converted to a sprite sheet for integration into Unreal (c) Projection card with sprite sheet applied inside Unreal Viewport

rendering times to review results, our approach ensures that all departments have instant access to the latest state of the shot. This immediacy dramatically accelerates the creative process, allowing for on-the-fly adjustments and ensuring that all team members are aligned with the evolving vision of the project. Our 'Shot is King' philosophy, supported by innovative tools and methodologies, establishes a new standard in animation workflows, bridging the gap between the artistic depth of traditional animation and the dynamic possibilities of real-time rendering.

3 ADDITIONAL CHALLENGES

In tackling the unique aesthetic demands of "Spice Frontier," Steamroller Animation has innovated beyond traditional frameworks, integrating advanced tools and processes within the Unreal Engine to enable the project's distinctive visual style. This section outlines the key technological advancements crafted to navigate the challenges of merging traditional 2D stylings with real-time rendering capabilities.

Central to our technical innovation is a custom dynamic character blueprint system, this is a solution born from the need to balance the flexibility of traditional CG animation with Unreal Engine's requirements for real-time efficiency (Figure 2b). This system manages multiple character versions critical for overcoming Unreal's skeleton compatibility and memory constraints. It uniquely empowers artists with the ability to directly influence the character's appearance, ensuring artistic decisions are preserved across version changes as well as optimizing memory usage.

Building on this foundation, we introduced a proprietary post-process Rim Shader. Unlike traditional methods that rely on a single post-process volume for scene-wide effects, our approach assigns

each character an individual post-process component. This allows for nuanced control over rim lighting effects, enabling artists to define distinct visual profiles for each character (Figure 3a). Such granular control is vital for achieving "Spice Frontier's" stylized 2D look. Alongside the rim shader we developed a custom gobo setup to help ground our characters in the world despite their 2D look. Through this setup, custom gobo actors can project shadow textures to specific scene actors. This method of shadow integration, controlled via Sequencer, not only enriches the visual depth but also maintains the project's artistic integrity.

The challenge of integrating traditional 2D FX into "Spice Frontier" prompted the creation of another unique toolset. By converting hand-drawn effects into optimized sprite sheets we can integrate them into Unreal (Figure 3b). This process ensures that the hand-drawn aesthetic is not lost in translation to the digital medium. Our proprietary tools facilitate the accurate alignment of these effects within 3D space, enabling dynamic integration without the need for a separate compositing stage (Figure 3c).

Together, these advancements reflect Steamroller Animation's dedication to pushing the boundaries of animation technology. By thoughtfully integrating these tools into the Unreal Engine, we not only leverage the technical prowess of Unreal Engine but also ensure the artistic integrity and stylistic ambitions of "Spice Frontier" are fully realized without compromise.

4 DISCUSSION

In this paper, we have detailed Steamroller Animation's pioneering approach to animation production for "Spice Frontier," utilizing Unreal Engine to fundamentally transform traditional shot-building techniques. This novel methodology, underpinned by our 'Shot is King' philosophy, has been augmented with a custom toolset designed to refine visual output and streamline workflow efficiencies. These innovations not only facilitate a more dynamic and integrated creative process but also significantly enhance the capacity for nuanced storytelling within a real-time rendering environment.

Looking ahead, we are committed to the continued development of our toolset, aiming to introduce even greater complexity and visual refinement into our workflows. A key area of focus will be the integration of production processes with the previsualization, layout, and storyboarding phases. By enhancing these early stages of production with our real-time systems, we aim to empower storytellers to make more accurate and impactful decisions from the outset.

In conclusion, the integration of Unreal Engine into our animation workflow represents a significant leap forward in the field. As we continue to refine and expand our toolset, we look forward to exploring how these advances can further revolutionize the animation process, making high-quality storytelling more accessible and efficient for creators everywhere.

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